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LESSON PLAN IN MATH 5

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Mr. Sunny Fernandez Teaching Internship Mentor	**Grade 5** Grade Level
December 1, 2025 Date of Submission	**8:00 AM - 9:50 PM** Time Schedule

Content Standard	*The learners should have knowledge and understanding of:* - prime and composite numbers.
Learning Competencies	*The learners shall be able to:* - distinguish prime numbers from composite numbers using the Sieve of Erastosthenes.

I. Intended Learning Outcomes (Objectives)

At the end of the lesson and through varied learning activities, the Grade 5 learners with at least 75% proficiency will be able to:

a. differentiate between prime and composite numbers;
b. demonstrate strict adherence to systematic procedures by completing the sieve of Erastothenes; and
c. use the Sieve of Eratosthenes to identify prime numbers up to 100.

II. Learning Content

A. **Topic (s)**	- **Prime and Composite Numbers**
B. **Key Concepts**	(1) prime numbers are numbers with only two factors, 1 and itself; and (2) composite numbers are numbers with 2 or more factors (excluding 1).
C. **References** *(Print, electronic, etc.)*	- Realistic Math Basic, Beyond, Breakthroughs - Grade 5. (2025). In *Phoenix Publishing House, Inc.* - (2nd ed.). - Pandamathfest. (2018, November 17). *GREECE: Sieve of Eratosthenes*. Panda Math Fest. https://pandamathfest.wordpress.com/2018/11/11/greece-sieve-of-eratosthenes/
D. **Instructional Materials**	- Laptop - Canva Presentation
E. **Integration** *(Values, SDG/ESD, GAD, etc.)*	- Quality Education (SDG 4): Reinforces mental math and multiplication mastery, which are vital prerequisites for mathematical literacy and lifelong learning. - Industry, Innovation, and Infrastructure (SDG 9): Develops algorithmic thinking and pattern recognition, which are the foundational logic skills used in computer science and data encryption. - Values Integration (Patience and Discipline): Instills the value of systematic effort and attention to detail, as students must carefully filter numbers to accurately complete the Sieve of Eratosthenes.
F. **Skills** *(21st century skills)*	- Critical Thinking and Problem Solving - Creativity and Innovation - Collaboration - Information, Media, and Technology Literacy
Preliminaries	- Prayer - Attendance Checking
III. Teaching and Learning Process (Experiences)	

I. Introduction (3 minutes)

The teachers will open the class by formally introducing themselves as the new teachers. They will provide a brief personal introduction and clearly state their expectations for the learners moving forward. This introduction sets the tone for the class and helps establish a positive and respectful learning environment before transitioning to the new lesson.

Elicit (5 minutes)

After the introduction, the teacher transitions to the elicitation phase by activating the learners' prior knowledge on **divisibility rules**.

Activity: Rule Recall

The teacher informs the class that before exploring the new lesson, they will first recall useful rules that help determine whether a number is divisible by another.

The teacher begins by presenting a short set of numbers on the screen and asks the students to identify which divisibility rules apply to each. The students quickly share their reasoning using the rules they recalled.

Students may respond by unmuting, typing in the chat, or raising their virtual hands. After several answers have been shared, the teacher proceeds with the recall questions to check understanding and reinforce correct rules:

1. What is the divisibility rule for 2?
   - Its last digit is even.
2. How do we check if a number is divisible by 5?
   - Its last digit is 0 or 5.
3. How do we know if a number is divisible by 3 or 9?
   - Divisible by 3: If the sum of all its digits is divisible by 3.
   - Divisible by 9: If the sum of all its digits is divisible by 9.
4. What is an example of a number divisible by 10, and why?

- Any number that ends with 0 is divisible by 10.

This elicitation activity allows the teacher to assess the learners' familiarity with divisibility rules and prepares them for identifying factor pairs and understanding prime and composite numbers in the upcoming lesson.

B. Engage (8 minutes)

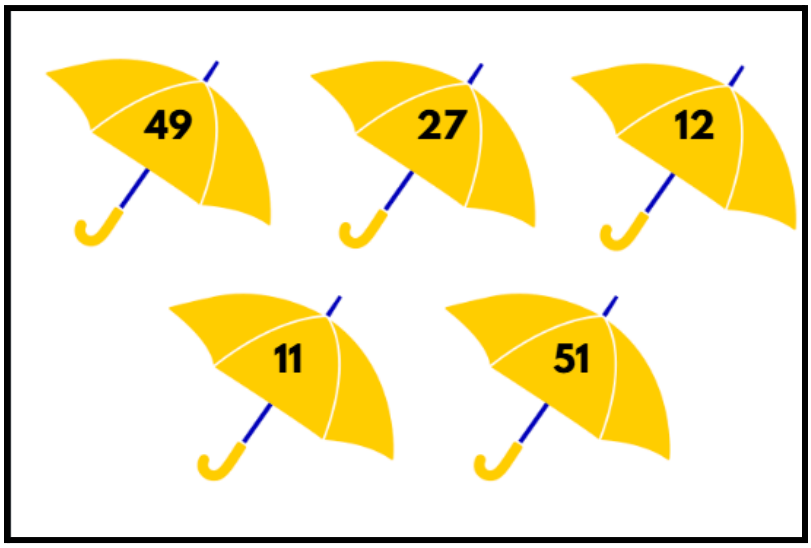
Building on the students' observations and responses from the elicitation phase, the teacher transitions to the engage activity.

Activity: Drop Match!

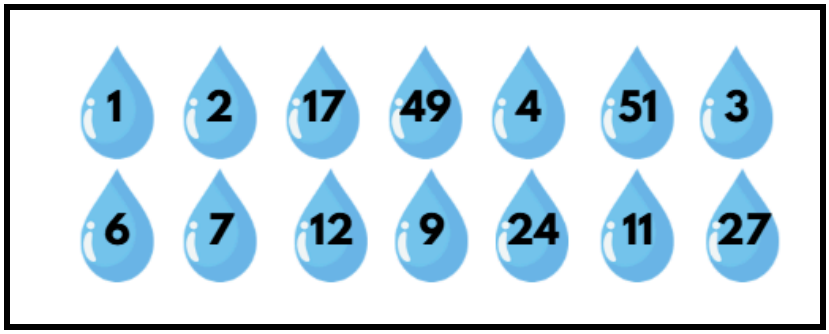
In this activity, the teacher displays the slide containing umbrellas with numbers and raindrops below, explaining that each umbrella represents a number and the raindrops show possible factors.

Students are instructed to carefully observe the numbers on the umbrellas. Each student selects one umbrella number to work on and writes it on their paper. They then examine the raindrops to identify and list all the factors of their chosen number.

A. Target Number



B. Factors



Once the students have completed their work, the teacher asks for volunteers to share their answers with the class, highlighting the umbrella number and its factors on-screen for discussion. Incorrect answers are addressed through probing questions to help students reason out their choices.

This activity allows the teacher to assess learners' prior understanding of factors and sets the stage for the formal discussion of prime and composite numbers.

C. Explore (15 minutes)

Prior to the lesson, the teacher will have the students prepare a worksheet with the numbers 1 - 100, arranged in a 10x10 box as seen in the figure below.

During the lesson, the teacher will verify that all students have their prepared 1-100 number grid worksheets ready before starting the lesson. The teacher will display a digital grid via screen share and guide the students step-by-step to black out 1, circle the prime numbers (2, 3, 5, 7), and eliminate their multiples. The students will follow the teacher's instructions in real-time, marking their individual worksheets at their stations. The teacher will give an example for the sieve from 1 - 10.

Upon completion of the grid, the teacher will pose the following inquiry series to the class:

1. What numbers have we circled?
   - 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97
2. Look at the numbers we circled. What is the one thing they all have in common?
   - They can only be divided by 1 and themselves.
3. We blacked out the number 1 at the very beginning instead of circling it or crossing. Why?"
   - It does not follow the pattern of both the circled, and crossed numbers.
4. In the Sieve, we had to follow a specific order (2, then 3, then 5...). What would happen to our results if we got impatient and skipped ahead?
   - We would miss some numbers (like 9 or 15) and mistakenly call them circle them, instead of crossing them out. It teaches us that skipping steps leads to errors.

The students will analyze their completed grids and provide answers via the meeting chat box or by unmuting their microphones.

	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

D. Explain

Based on the activity, the students will be asked to define:

- **Prime Numbers:**
  - Numbers that have only TWO factors: 1 and themselves;
  - *Examples:* 2, 3, 5, 7, 11...
- **Composite Numbers:**
  - Numbers that have 2 or more factors (excluding 1)
  - *Examples:* 4, 6, 8, 9…

How are prime numbers connected to composite numbers? - Prime numbers make up composite numbers

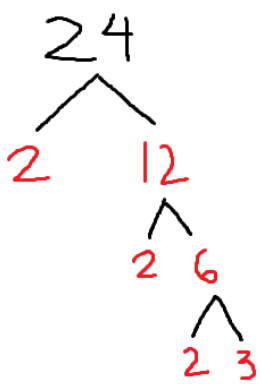
The students will be shown a sample of prime factorization:

Is it possible to break down 8 into 2 smaller numbers? - YES

Is there anything else we can break down? - YES, 4

We can say that the prime factorization of 8 is 2 x 2 x 2

Now try it with **24**

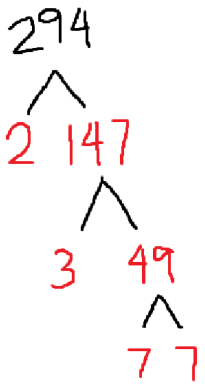


24 = 2 x 2 x 2 x 3

What about **89?**

89 = 1 x 89

Let's try with the 3 digits of **294?**



294 = 2 x 3 x 7 x 7

If a number can be broken down into **2 or more** branches, it is a **Composite Number**

If a number can **ONLY** be broken down into **2** branches (1 and itself), it is a **Prime Number**

E. Elaborate (20 minutes)

Activity: Find the Factors

In this activity, the students will work individually to apply their understanding of prime factorization and the classification of numbers as prime or composite. Each learner will be given a set of numbers. For every number, the students will list all its prime factors by using the Prime Factorization method.

After identifying the factors, the students will determine whether each number is prime or composite and write their answers accordingly, as well as show their solutions.

1. (59) Factors: 1 x 59

   Classification: PRIME

2. (67) Factors: 1 x 67

   Classification: PRIME

3. (69) Factors: 3 x 23

   Classification: COMPOSITE

4. (420) Factors: 2 x 2 x 3 x 5 x 7

Classification: COMPOSITE

5. (119) Factors: 7 x 17

Classification: COMPOSITE

Following the factorization exercise, the students will answer a set of problem-solving questions related to prime and composite numbers.

PROBLEM SOLVING

1. If you multiply the first three odd prime numbers together, what is the resulting product?

Answer: 105

2. The sum of two distinct prime numbers is 39. When you subtract the smaller prime from the larger prime, the result is a composite number. What is the product of the two prime numbers?

Answer: 74

3. A secret code is a prime number between 10 and 20. If you reverse its digits, the resulting number is also prime. If the sum of its digits is a perfect cube, what is the secret code?

Answer: 17

Challenge: The sum of three distinct positive integers, A, B, and C, is 18. If A is a prime number, B is a composite number, and C is a number that is neither prime nor composite, what is the product of A x B x C, such that the sum of the prime factors of the product is equal to 17?

Answer: 52

IV. Learning Assessment (Evaluate)

The teacher informs the learners that they will complete an asynchronous Google Forms quiz on prime and composite numbers. The teacher posts the quiz link and instructions on the online platform. Students open the link on their own time, enter their name, and answer the quiz independently.

Prime and Composite Numbers – Quiz
Name: _____ Grade/Section: _____ Score: _____/10

Direction: Read each question carefully. Choose the letter of the correct answer.

1. Which of the following numbers is a prime number?
    a. 9
    b. 10
    c. 11
    d. 12

2. How many factors does 1 have?
    a. 0
    b. 1
    c. 2
    d. 4

3. What is the smallest composite number?
    a. 2
    b. 4
    c. 3
    d. 5

4. If a number has only 2 factors, it's always a prime number. True or False?
    a. True, prime numbers always have 2 factors.
    b. False, not all numbers with only 2 factors are prime.
    c. True, there exists no composite number that only has 2 factors.
    d. False, prime numbers do not have only 2 factors.

5. What is the prime factorization of 1024?
    a. 2 x 2 x 2 x 2 x 2 x 2 x 2 x 2 x 2 x 2
    b. 2 x 2 x 2 x 2 x 2 x 2 x 2 x 2 x 2
    c. 2 x 2 x 2 x 2 x 2 x 2 x 2
    d. 2 x 2 x 2 x 2 x 2 x 2 x 2 x 2

6. A mystery number has exactly three prime factors. Two of the factors are 2 and 3. The third factor is a prime number between 10 and 20. Which of the following could not be the mystery number?
    a. 66
    b. 102
    c. 78
    d. 90

7. Analyze the prime factorization of two numbers: Number A: 2 x 3 x 5, Number B: 2 x 2 x 3 x 5 How many times larger is Number B than Number A?
    a. 2 times larger
    b. 3 times larger
    c. 5 times larger
    d. They are the same.

8. What is the smallest number with 3 distinct prime factors?
    a. 24
    b. 12
    c. 30
    d. 6

9. Are all even numbers composite numbers?
    a. Yes, all even numbers are divisible by 2.
    b. No, all composite numbers are even.
    c. Yes, all even numbers are crossed out in the sieve of Eratosthenes.
    d. No, 2 is only divisible by 2 and 1.

10. Which of the following numbers is not part of the group?
    a. 47
    b. 67
    c. 51
    d. 43

V. Assignment (Extend)

Assignment: "The Super Sieve"

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100
101	102	103	104	105	106	107	108	109	110
111	112	113	114	115	116	117	118	119	120
121	122	123	124	125	126	127	128	129	130
131	132	133	134	135	136	137	138	139	140
141	142	143	144	145	146	147	148	149	150
151	152	153	154	155	156	157	158	159	160
161	162	163	164	165	166	167	168	169	170
171	172	173	174	175	176	177	178	179	180
181	182	183	184	185	186	187	188	189	190
191	192	193	194	195	196	197	198	199	200

During the Extend activity, the teacher explains that the class will now apply the Sieve of Eratosthenes to the numbers 1–200. The teacher gives each learner the Super Sieve sheet and reminds them to circle primes, cross out multiples, and color primes green and composites yellow. The Grade 5 students work individually. After completing the sieve, students count their primes and composites and write a short three-sentence reflection about what they learned.

Specifically, this will be the task instructions to be given to the students:

1. Get your activity sheet with the numbers 1 to 200 written in a grid.
2. Ignore number1 because it is not prime and not composite.\
3. Find the first prime number, 2.
    a. Circle 2.
    b. Then cross out all multiples of 2 (4, 6, 8, 10, …) up to 200.
4. Go to the next uncrossed number, 3.
    a. Circle 3.
    b. Cross out all the multiples of 3.
5. Continue the pattern using the next uncrossed number each time (5, 7, 11, …) until you finish sieving up to 200.
6. When you are done, count how many prime numbers you found from 1–200.
7. Color all prime numbers in **green**, and color all composite numbers in **yellow**.
8. After that, students will answer the reflection question, with a maximum of three sentences.
9. Submit your finished Super Sieve Chart into the Google Classroom

Processing Questions:

1. **How many prime numbers are there?**
   -46
2. **How many composite numbers are there?**
   -153

B.

Reflection:

1. **What did you learn from doing the Super Sieve activity?**

- Answers may vary

Google Document:

https://docs.google.com/document/d/18i6irD61RQAeSS7Y2FXeR4A7xoZWUUqop0ijQk6iZcc/edit?usp=sharing

Rubric for Assignment: "The Super Sieve"

Criteria	5 – Excellent	4 – Proficient	3 – Developing	2 –Beginning
Accuracy of Identifying Prime & Composite Numbers	All numbers are correctly classified; no errors in circling, crossing out, or coloring.	1–3 minor errors in identifying primes or composites.	Several errors (4–7); shows partial understanding..	Many errors; most numbers incorrectly classified.
Completeness of the Super Sieve (1–200)	The entire 1–200 grid is finished with all primes circled and all composites crossed out; all colors are applied.	Mostly complete; only 1–10 numbers not processed or missing color.	Partially complete; 11–25 numbers not processed or missing color.	Incomplete; more than 25 numbers missing or no clear work shown.
Neatness and Organization	Work is very neat, readable, and well-organized; colors and markings are clear.	Work is readable with minor untidiness.	Some parts are messy or unclear, but still understandable.	Work is difficult to read or very disorganized.
Reflection Phase				
Reflection Understanding & Connection to Real Life	Reflection clearly explains the usefulness of prime and composite numbers with specific examples from real life; shows deep understanding.	Reflection explains usefulness with some examples; understanding is evident but not fully detailed.	Reflection shows limited understanding; examples are vague or unclear..	Reflection is missing, unclear, or shows no understanding..
Effort & Presentation Clarity & Effort	3 well-written sentences; ideas are logical, neat, and easy to read; shows strong effort.	2 sentences; mostly clear and organized; effort is evident.	1 sentence; somewhat unclear or disorganized; minimal effort.	Missing; ideas are confusing or poorly presented; little to no effort.

Scoring Guide:

- **8-10 points = Outstanding**
- **5-7 points = Very Good**
- **2-4 points = Needs Improvement**
- **Below 1 points = Incomplete**